

## 1.7 Proof by induction

### Further Pure 1 · 2026 examination · Concept guide and original exam-style practice

#### Understand the concept

Induction proves a proposition for all positive integers by establishing a base case and showing  $P(k)$  implies  $P(k+1)$ . The inductive hypothesis is used at a visible algebraic step, not merely asserted. For divisibility, rewrite the  $n=k+1$  expression as a known divisible term plus a clearly divisible remainder.

#### Worked example: follow the reasoning

If  $\sum r$  from 1 to  $n = n(n+1)/2$ , adding  $n+1$  to the assumed  $k$ -sum yields  $(k+1)(k+2)/2$ .

#### Structured practice

Answer in the space provided. Show working and units where relevant.

1 Prove  $1+3+\dots+(2n-1)=n^2$  by induction.

[3]

Final answer: \_\_\_\_\_

# Answers and explanation

## Question 1 [3 marks]

$n=1$ :  $1=1^2$  [1]. Assume sum through  $k$  is  $k^2$  [1]; add  $2(k+1)-1=2k+1$  to get  $k^2+2k+1=(k+1)^2$  [1].

### Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Further Mathematics 9231, 2026–2027.

<https://www.cambridgeinternational.org/Images/697357-2026-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.